

Task 1

Embedded Systems & Internet of Things (IoT)

1. Introduction to Embedded Systems

What is an Embedded System?

An Embedded System is a special-purpose computer system designed to perform one specific task or a limited number of tasks. Unlike a general-purpose computer such as a desktop or laptop, which can run many different applications, an embedded system is programmed to perform a dedicated function with high efficiency, accuracy, and reliability. An embedded system consists of both hardware and software. The hardware includes components such as a microcontroller or microprocessor, memory, sensors, actuators, and input/output devices, while the software (called firmware) controls the operation of the hardware.

The main purpose of an embedded system is to monitor, control, or automate a particular process. Most embedded systems operate in real time, meaning they respond immediately to inputs within a fixed time limit. Because of this, they are widely used in industries where fast and accurate responses are essential.

Today, embedded systems are found in almost every electronic device we use in daily life. Modern technologies such as the Internet of Things (IoT), smart homes, healthcare devices, robotics, industrial automation, and automotive systems rely heavily on embedded systems.

Characteristics of Embedded Systems

- Designed for a specific application.
- Performs tasks automatically with minimal human intervention.
- Operates in real time.
- Compact in size and consumes low power.
- Highly reliable and efficient.
- Combines hardware and software into a single system.
- Cost-effective and easy to integrate into electronic products.

Applications of Embedded Systems

Embedded systems are used in almost every sector because they provide automation, accuracy, and fast response.

I. Smart Home Systems

Smart homes use embedded systems to automate household devices.

Examples:

- Automatic lighting systems
- Smart door locks
- Smart fans and air conditioners
- Home security systems
- Fire and smoke detection systems

Benefits:

- Saves electricity
- Improves security
- Provides remote control using mobile applications

II. Automotive Systems

Modern vehicles contain dozens of embedded systems that improve safety, comfort, and performance.

Examples:

- Airbag control system
- Anti-lock Braking System (ABS)
- Engine Control Unit (ECU)
- Cruise control
- Parking assistance
- GPS navigation system

Benefits:

- Increases passenger safety
- Improves fuel efficiency
- Reduces accidents

III. Healthcare Devices

Medical equipment uses embedded systems for accurate monitoring and diagnosis.

Examples:

- Digital thermometer
- Blood pressure monitor
- ECG machine
- Insulin pump
- Heart pacemaker
- Pulse oximeter

Benefits:

- Accurate measurements
- Continuous patient monitoring
- Faster medical diagnosis

IV. Industrial Automation

Factories use embedded systems to automate machines and production processes.

Examples:

- PLC-based automation
- Conveyor belt control
- Temperature monitoring
- Robot arms
- Quality inspection systems

Benefits:

- Increases production
- Reduces human errors
- Improves product quality

V. Robotics

Robots use embedded systems for sensing, controlling motors, and decision-making.

Examples:

- Industrial robots
- Medical robots
- Delivery robots
- Educational robots

Benefits:

- High precision
- Faster operations
- Reduced human effort

VI. Internet of Things (IoT)

IoT devices combine embedded systems with the internet to exchange data.

Examples:

- Smart agriculture
- Smart energy meter
- Weather monitoring system
- Smart parking
- Smart irrigation

Benefits:

- Remote monitoring
- Data collection
- Improved efficiency

Microcontroller vs Microprocessor

Although both microcontrollers and microprocessors are electronic computing devices, they are designed for different purposes.

Microcontroller	Microprocessor
Designed for specific tasks	Designed for general-purpose computing
CPU, RAM, ROM, and I/O ports are integrated on a single chip	Only CPU is present on the chip
Requires fewer external components	Requires external RAM, ROM, and I/O devices
Low power consumption	Higher power consumption
Lower cost	Higher cost
Used in embedded systems	Used in computers and laptops
Smaller size	Larger system size
Performs one dedicated task	Can perform multiple complex tasks

2. Component Study

2.1 Arduino UNO

Introduction

Arduino UNO is one of the most popular microcontroller development boards used in embedded systems and IoT projects. It is based on the ATmega328P microcontroller developed by Atmel (now Microchip Technology). Arduino UNO is beginner-friendly because it is easy to program using the Arduino IDE and supports a wide range of sensors and modules.

The board contains digital and analog input/output pins, a USB interface, power supply connectors, and communication interfaces. It is widely used for learning electronics, automation, robotics, and smart device development.

Features

- Based on ATmega328P microcontroller
- Operating Voltage: 5V
- Input Voltage: 7–12V
- 14 Digital I/O Pins (6 PWM)
- 6 Analog Input Pins
- 32 KB Flash Memory
- 2 KB SRAM
- 1 KB EEPROM
- Clock Speed: 16 MHz
- USB interface for programming

Advantages

- Easy to use
- Low cost
- Large community support
- Compatible with many sensors and modules

Applications

- Smart home automation
- Robotics
- Temperature monitoring
- Smart lighting systems
- IoT projects
- Industrial automation

2.2 ESP32

Introduction

ESP32 is a powerful 32-bit microcontroller developed by Espressif Systems. It has built-in Wi-Fi and Bluetooth, making it ideal for Internet of Things (IoT) applications.

Unlike Arduino UNO, ESP32 can connect directly to the internet and communicate with cloud platforms, mobile applications, and web servers.

Features

- Dual-Core Processor (240 MHz)
- Built-in Wi-Fi
- Built-in Bluetooth
- Operating Voltage: 3.3V
- Multiple GPIO Pins
- Analog and Digital Inputs
- PWM Support
- Low Power Consumption

Advantages

- High processing speed
- Wireless communication
- Suitable for IoT applications
- Supports Bluetooth devices

Applications

- Smart Home Systems
- Smart Agriculture
- Wireless Sensor Networks
- Weather Monitoring
- Home Automation
- IoT Devices

2.3 Raspberry Pi Pico

Introduction

Raspberry Pi Pico is a low-cost microcontroller board developed by the Raspberry Pi Foundation. It is based on the RP2040 microcontroller, which offers excellent performance with low power consumption.

The Pico supports programming in C/C++ and MicroPython, making it suitable for education and embedded system projects.

Features

- RP2040 Dual-Core ARM Cortex-M0+
- Clock Speed: 133 MHz
- Operating Voltage: 3.3V
- 26 GPIO Pins
- ADC Support
- PWM Support
- USB Programming
- Low Power Consumption

Advantages

- Low cost
- High speed
- Easy programming

Applications

- Embedded Systems
- Robotics
- Automation
- Sensor Interfacing
- IoT Projects
- Data Acquisition Systems

2.4 Temperature Sensor

Introduction

A Temperature Sensor is an electronic device that measures the surrounding temperature and converts it into an electrical signal. The microcontroller reads this signal and performs the required action.

One of the most commonly used sensors with Arduino is the LM35 Temperature Sensor.

Working Principle

The LM35 measures temperature and produces an analog output voltage. The output voltage increases linearly with temperature. Arduino reads this voltage through an analog input pin and converts it into temperature in degrees Celsius.

Features

- Operating Voltage: 4V–30V
- Output: Analog Voltage
- High Accuracy
- Low Power Consumption

Applications

- Room Temperature Monitoring
- Air Conditioning Systems
- Medical Equipment

- Industrial Temperature Control
- Weather Stations

Advantages

- Easy to interface
- Accurate readings
- Low cost
- Reliable operation

2.5 Motion Sensor (PIR Sensor)

Introduction

A Motion Sensor detects the movement of people or objects. The most commonly used motion sensor is the PIR (Passive Infrared) Sensor.

It detects infrared radiation emitted by the human body and sends a digital signal to the microcontroller whenever motion is detected.

Working Principle

When a person enters the sensing area, the PIR sensor detects the change in infrared energy and sends a HIGH signal to the Arduino. The Arduino then turns ON an LED, buzzer, or alarm.

Features

- Operating Voltage: 5V
- Digital Output
- Motion Detection Range: 3–7 meters
- Low Power Consumption
- Easy to Interface

Applications

- Security Systems
- Automatic Doors
- Motion Detection Alarm

- Smart Lighting
- Home Automation

Advantages

- Low power consumption
- High reliability
- Easy installation
- Long sensing distance

2.6 Light Sensor (LDR Sensor)

Introduction

An LDR (Light Dependent Resistor) is a sensor that detects the intensity of light. Its resistance changes according to the amount of light falling on it.

The LDR is widely used in automatic lighting systems.

Working Principle

- Bright Light: Resistance decreases.
- Darkness: Resistance increases.

Arduino reads the voltage across the LDR using an analog pin. If the light level is low, Arduino turns ON the LED. When sufficient light is available, the LED turns OFF.

Features

- Low Cost
- Analog Output
- High Sensitivity
- Easy to Interface
- Low Power Consumption

Applications

- Automatic Street Lights

- Garden Lighting
- Smart Home Lighting
- Solar Tracking Systems
- Camera Light Meters

Advantages

- Simple design
- Reliable performance
- Energy saving
- Low maintenance

3. Mini Project: Smart Lighting System Using Arduino UNO and LDR Sensor

Project Title: Smart Lighting System Using Arduino UNO and LDR Sensor

I. Objective

The objective of this project is to design and develop a Smart Lighting System that automatically switches ON the light in darkness and switches it OFF when sufficient light is available. The system uses an LDR (Light Dependent Resistor) to detect light intensity and an Arduino UNO to control an LED.

This project demonstrates the basic concept of Embedded Systems by combining a sensor, a microcontroller, and an output device to perform automatic control.

II. Components Required

Sr. No.	Component	Quantity
1	Arduino UNO	1
2	LDR (Light Sensor)	1
3	LED	1
4	220 Ω Resistor	1
5	10k Ω Resistor	1
6	Breadboard	1
7	Jumper Wires	As required
8	USB Cable	1

III. Working Principle

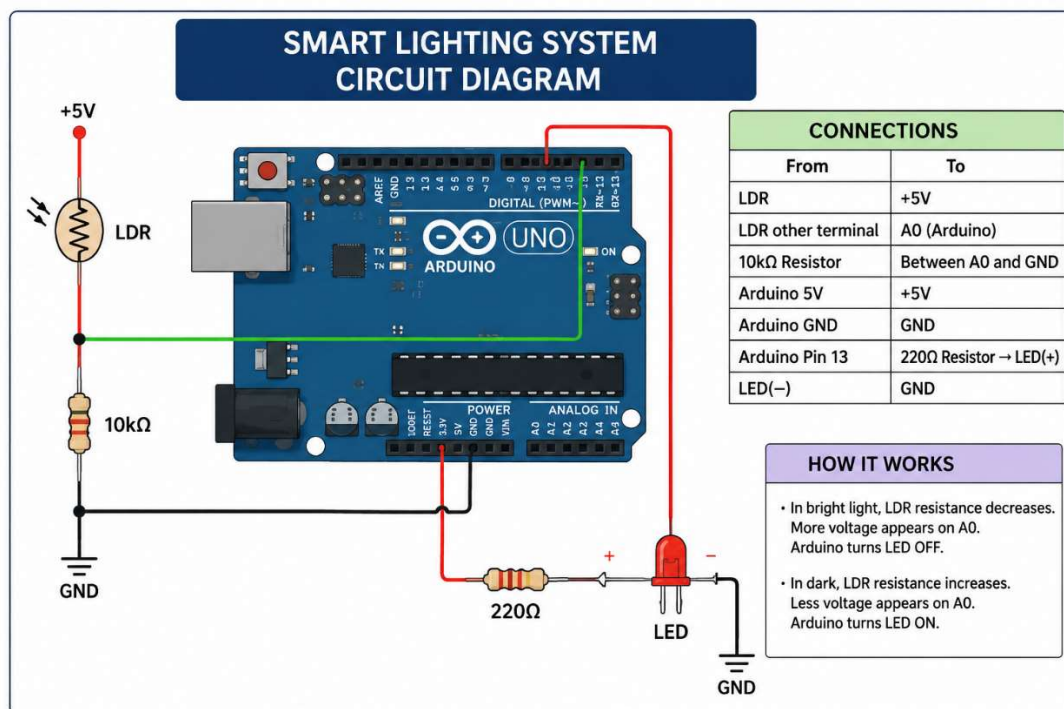
The Smart Lighting System works on the principle of light intensity detection using an LDR sensor. An LDR (Light Dependent Resistor) changes its resistance according to the amount of light falling on it. During daytime, the LDR receives more light, its resistance decreases, and Arduino detects a higher light level. The LED remains OFF.

During nighttime or in darkness, the LDR receives less light, its resistance increases, and Arduino detects a lower light level. The LED turns ON automatically. The Arduino continuously reads the analog value from the LDR sensor and compares it with a predefined threshold value. Based on this comparison, it controls the LED.

Working Steps

- Power ON the Arduino UNO.
- LDR senses the surrounding light intensity.
- Arduino reads the analog value from the LDR.
- If the light intensity is below the threshold value, Arduino switches the LED ON.
- If the light intensity is above the threshold value, Arduino switches the LED OFF.
- The process repeats continuously.

IV. Circuit Diagram



V. Arduino Code

```
int LDR = A0;
int LED = 13;

void setup()
{
  pinMode(LED, OUTPUT);
}
```

```

void loop()
{
  int sensorValue = analogRead(LDR);

  if(sensorValue < 500)
  {
    digitalWrite(LED, HIGH);
  }
  else
  {
    digitalWrite(LED, LOW);
  }

  delay(100);
}

```

VI. Expected Output

- Bright Light (Day): LED remains OFF.
- Low Light (Night): LED turns ON automatically.

VII. Applications of Smart Lighting System

The Smart Lighting System is an embedded system project that automatically controls lights based on the surrounding light intensity. It uses an LDR (Light Dependent Resistor) sensor to detect the amount of light and an Arduino UNO to control the LED or lamp. This project has many real-life applications because it helps save electricity, reduces human effort, and improves convenience.

Automatic Street Lighting

One of the most common applications of the Smart Lighting System is in street lighting. During the daytime, the LDR sensor detects sunlight and keeps the street lights OFF. When it becomes dark in the evening, the sensor detects low light intensity, and the Arduino automatically switches the lights ON. This reduces electricity consumption and eliminates the need for manual operation.

Home Automation

Smart lighting systems are widely used in homes. Lights can automatically turn ON when the room becomes dark and turn OFF when there is sufficient natural light. This improves comfort and reduces unnecessary power consumption.

Garden and Outdoor Lighting

Gardens, parks, and pathways often use automatic lighting systems. The lights remain OFF during the day and automatically turn ON at night, providing safety and improving the appearance of outdoor areas.

Parking Areas

Parking lots require proper lighting for vehicle movement and security. A smart lighting system ensures that lights operate only when needed, reducing energy costs and improving visibility at night.

Offices and Commercial Buildings

In offices and commercial buildings, smart lighting systems automatically adjust lighting according to natural daylight. This creates a comfortable working environment while reducing electricity bills.

Industrial Automation

Industries use automatic lighting in warehouses, factories, and production areas. Embedded systems help control lighting efficiently, improve worker safety, and reduce operational costs.

Educational Institutions

Schools, colleges, and universities can install smart lighting systems in classrooms, libraries, and corridors. The lights operate automatically based on ambient light, helping institutions save electricity.

Smart Cities

Smart lighting is an important part of smart city projects. Intelligent street lighting systems improve public safety, reduce maintenance costs, and increase energy efficiency.

VIII. Advantages of the Smart Lighting System

- Saves electrical energy.
- Reduces electricity bills.
- Automatic operation without human intervention.

- Environment-friendly solution.
- Simple and low-cost design.
- Reliable and easy to maintain.

IX. Conclusion

The Smart Lighting System is a simple and effective embedded system project that demonstrates the practical use of sensors and microcontrollers. In this project, an LDR sensor continuously measures the surrounding light intensity, while the Arduino UNO processes the sensor data and automatically controls the LED or light source.

The project helps in saving electrical energy, reducing manual effort, and improving automation. It also introduces the basic concepts of embedded systems such as sensor interfacing, microcontroller programming, input/output control, and real-time operation.